

ET AI HACKATHON 2026

Problem Statements

Industrial | Smart Cities | Sustainability | Supply Chain | Geospatial | Public Safety | Sales Excellence

Each team must select one problem statement. Choose the challenge that best aligns with your team's skills, expertise, and interests.

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AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

Theme: *Industrial Intelligence / Worker Safety / Geospatial Safety Analytics*

PROBLEM CONTEXT

India's heavy industrial sector continues to pay a devastating human cost. According to DGFASLI, over 6,500 fatal workplace accidents were recorded in FY2023 — and that figure excludes most of the mining and construction sectors. In one of the most disturbing recent incidents, eight workers died at Visakhapatnam Steel Plant in January 2025 when entrapped gases triggered a sudden explosion in the coke oven battery — a facility that had functioning safety systems including gas detectors, permit-to-work controls, and SCADA. An investigation by The Wire found that warning signals from gas pressure sensors existed, but no intelligence layer connected those readings to operational decisions in time. This pattern — data present, but unacted upon — repeats itself across Indian heavy industry. A FICCI survey in 2024 found that over 60% of large industrial facilities rely on manual handoffs to coordinate between their own digital safety tools. The problem is not the absence of technology. It is the absence of a unified intelligence layer that fuses data from disparate sensors, shift logs, maintenance records, and video feeds into a real-time risk picture — and acts on it before a fatality, not after.

CHALLENGE STATEMENT

Build an AI-powered Industrial Safety Intelligence platform that brings together data from IoT sensors, SCADA systems, permit-to-work logs, CCTV feeds, and shift records into a single predictive layer. The system should detect compound risk conditions — like the co-occurrence of maintenance activity and hazardous gas accumulation — that no single sensor would flag alone, and trigger preemptive interventions before they escalate.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Compound Risk Detection Engine** — Multi-agent system that correlates gas sensor readings, work permit activity, equipment maintenance status, and shift changeover patterns to identify dangerous combinations — such as confined space entry during abnormal process conditions — hours before they become critical.
- **Geospatial Safety Heatmap** — Real-time geospatial layer over plant layout that visualises risk zones dynamically as conditions change — integrating worker location data, hazardous area classifications, and active permit overlaps to give safety officers situational awareness across the entire facility.

- **Incident Pattern Intelligence** — RAG-powered agent that cross-references near-miss reports, historical incident data, and OISD/Factory Act regulatory guidance to identify recurring patterns that manual investigations miss — and surfaces them as actionable prevention priorities.
- **Digital Permit Intelligence Agent** — AI that analyses active permits against real-time plant conditions and flags dangerous simultaneous operations — for example, hot work permits issued in proximity to areas with elevated gas readings, a combination that has repeatedly preceded fatal incidents.
- **Emergency Response Orchestrator** — Autonomous agent that, on confirmed trigger, immediately initiates evacuation protocols, alerts response teams across channels, preserves sensor evidence, and generates a preliminary regulatory-compliant incident report — reducing the critical first 10 minutes from chaos to coordinated response.
- **Quality & Compliance Audit Agent** — AI layer that continuously monitors safety procedures, inspection records, and statutory compliance documentation against regulatory standards (OISD, DGMS, Factory Act) — flagging deviations before audits and generating corrective action workflows automatically.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- Geospatial Intelligence & Plant Layout Analytics
- RAG over incident and regulatory document corpora
- Computer Vision & CCTV Analytics
- IoT / SCADA Data Integration
- Knowledge Graphs (equipment-permit-risk relationships)

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Compound risk detection accuracy versus single-sensor baselines, prediction lead time before incident threshold, geospatial evidence quality, regulatory compliance coverage (OISD/Factory Act/DGMS), and demonstrated reduction in false negative rate — the metric that actually saves lives.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%